

Friends in Higher Places: Social Fit and University Choice*

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Abstract

How does lack of social exposure to elite university students inhibit applications from talented low-SES high schoolers? We combine UK administrative data with a field experiment to show that perceptions of the social environment at elite universities are a central barrier. We exploit “breakthrough” events when a school first sends a student to a top university and find that applications and attendance to that university subsequently rise. Surveys show that low-SES students are pessimistic about their chances of fitting in socially at elite universities, but not their chances of receiving an offer. Random assignment to videos of undergraduates discussing their experiences and one-on-one mentorship increases applications to the treatment university by 14 percentage points. Student beliefs about their social fit at that university update positively, with no change in other beliefs. (*JEL* D71, D91, I23, I24, I26, Z13)

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I. Introduction

Elite universities are widely regarded as vehicles for social mobility. Low-SES students who graduate from elite universities see labor market returns on par with those of high-SES students (Chetty et al. 2020; Van Der Erve, Drayton, and Britton 2021). Yet socioeconomically disadvantaged students are substantially less likely to attend elite universities in the UK (Campbell et al. 2022), the USA (Chetty, Deming, and Friedman 2026), Norway (Cattan, Salvanes, and Tominey 2025), and Chile (Barrios-Fernández 2022), even conditional on high-school grades. In the UK, the setting of this paper, socioeconomic gaps persist despite uniform tuition fees across universities and income-based affirmative action. The disparity appears in university systems that differ widely, which points to causes beyond any single system’s design.

A natural candidate for why academically qualified low-SES students apply to elite universities at lower rates is differences in *social exposure*. Low-SES students are less likely to encounter individuals who attend elite universities within their social network, a channel that has been shown to matter in university enrollment decisions (Altmejd et al. 2021; Barrios-Fernández 2022; Bechichi and Kenedi 2024). Yet we know relatively little about *why* students respond to their peers’ university choices and which information frictions, if any, social exposure alleviates.

This paper examines how social exposure shapes beliefs about the university experience which in turn generate socioeconomic disparities in elite university applications. We conduct two complementary studies using UK administrative data and a high-stakes field experiment. In the first, we study the effects of quasi-random variation in exposure to elite universities on university and labor market outcomes. Our field experiment randomizes social exposure through videos and one-on-one mentorship with university students on a sample 640 university applicants across 21 high schools in England and Wales. In both studies, we find that application decisions are highly responsive to social exposure. These effects are largest among low-SES students, amplified by demographic similarity, and mediated through an upward shift in beliefs about social fit at elite universities.

We begin by documenting socioeconomic gaps in both applications and social exposure to elite universities in the UK. Low-income students are less likely to attend one of the 10 most academically selective UK universities and around 40% of this enrollment gap can be attributed to disparities in applications. At the same time, low-income students have substantially less social exposure to top universities through their high schools. Low-income students are 5–10 percentage points more likely to attend high schools where *no student* has attended any of the top 10 universities in the preceding three years.

We develop a simple framework to illustrate how systematic lack of exposure to elite universities can generate persistent pessimistic beliefs and uncertainty about elite universities. Students learn from past cohorts of schoolmates about the payoffs at an elite university, but learn the exact payoff only if a past schoolmate attended the elite university. Under an assumption of naïve social learning (Eyster and Rabin 2010), students infer that non-attendees at the university had low payoffs and thus revise their expected payoffs downward if their older schoolmates did not attend. This can lead to herding patterns, where students from schools with no applications to elite universities are persistently less likely to apply. However, a student with an idiosyncratically positive expected payoff may still choose to attend the elite university. This first student to attend the elite university can persistently raise applications among subsequent cohorts at that school.

Next, we show that application choices respond to quasi-random exposure to elite universities using UK administrative data. Specifically, we study *breakthroughs* to a university—cases where a student attends a particular university for the first time after several years in which no student from their school attended. In an event study framework, we find that breakthroughs persistently encourage applications to the specific university where the breakthrough occurs, raising application rates by around 30–50%. Breakthroughs have no effect on applications to other similarly ranked universities: breakthroughs to Cambridge increase applications to Cambridge among subsequent cohorts with no impact on applications to Oxford and vice versa. This suggests that low-SES students may face information frictions about *match-specific* payoffs for a particular university, rather than components that are transferable across universities or across all students. We also find that breakthrough effects are strongest among the students who are demographically *similar* to the breakthrough student.

Students shifted into elite universities graduate successfully and see substantial labor market returns, suggesting that these students are well-matched to these universities. Enrollment increases by 60–85% of the increase in applications and almost all of these students graduate successfully and on time. Using a matched-control design, we show that students who apply to an elite university following a breakthrough earn around £4,000 (US\$5,120) more per year than matched controls by age 27.

Why did these students forgo the large benefits to attending an elite university *prior* to the breakthrough, and what changed *after* the breakthrough? Our findings in the administrative data are consistent with the alleviation of some form of information friction upon a breakthrough, but cannot isolate the specific aspect of universities about which students lacked information. For example, low-SES students may be systematically misinformed about admissions requirements or the returns to university. Another potential explanation

behind the breakthrough effects is that students feel more comfortable attending a university at which they *already have a friend*. Moreover, larger effects among demographically similar students may simply reflect the fact those students were more likely to be friends and interact, rather than learning about a demographic-group specific aspect of the university experience. Understanding how demographic similarity mediates social learning in this context and identifying which beliefs are shifted is essential for designing policies to address socioeconomic disparities in higher education.

To distinguish between potential mechanisms, we conduct a high-stakes field experiment that randomizes social exposure to elite universities, varying the demographic overlap along with whether the exposure is passive (video clips) or relational (one-on-one mentorship). Our experimental sample consists of 640 high school students in their final year who expressed some interest in applying to universities in January 2026. All participants attend an in-school workshop that provides information about the application process as well as admissions requirements and data on average postgraduate earnings at various universities. Participants randomized into the treatment arm additionally watch two video clips of university students discussing their experiences. Treated participants are subsequently randomized into one of two treatment subarms for one-on-one mentorship with university students: T1 connects participants to demographically *similar* mentors and T2 connects participants to demographically *dissimilar* mentors.

The design of our experiment allows us to measure whether applications respond to social exposure, how this response varies with demographic similarity, and which beliefs shift in response to exposure. We ask participant to report their perceived likelihood that they would (a) get an offer from, (b) make friends and fit in at, or (c) graduate successfully from specific universities. By eliciting beliefs across baseline, midline, and endline surveys, we document how each belief evolves over the course of the experiment. We collect realized application choices directly from participating schools at the end of the experiment. The active control arm ensures that any estimated treatment effects reflect the impact of social exposure over and above that of acquiring admissions and earnings information.

Baseline surveys indicate that low-SES students are pessimistic about their probability of fitting in and making friends at elite universities, but not about their chances of receiving an offer or graduating successfully. Both low-income and first-generation students report perceived probabilities of social fit that are 6-8 percentage points lower than high-SES students with similar coursework and test score. Furthermore, we find that low-SES students hold beliefs about admissions requirements at elite universities that are correctly calibrated. Our findings suggest that a baseline lack of exposure can generate pessimism about match-specific aspects of elite universities without distorting beliefs about more

general aspects such as admissions requirements.

Both video and mentor exposure substantially shift participants' application choices. Pooling across treatments, assignment to treatment increases the probability of applying to one of the universities that the mentor or video speaker attended by 14 percentage points ($p = 0.004$). Treatment effects in our experiment are concentrated among low-SES students. We find a 17 percentage point ($p = 0.007$) increase in the probability of applying to the treatment university for low-SES students, while the effect is substantially smaller and insignificant for high-SES students ($b = 0.089, p = 0.258$). Video clips and mentorship are equally effective in inducing applications, suggesting that the effects of social exposure on elite university applications operate through the provision of new information about elite universities rather than friends at elite universities.

We find that students' application choices respond substantially more to mentors who are demographically similar to them. A similar mentor raises the probability of applying to the mentor's university by 21 percentage points ($p = 0.001$), while a dissimilar mentor has no effect. We do not find substantial differences in take-up rates between T1 (demographically dissimilar mentors) and T2 (demographically similar mentors). This clarifies the interpretation of the heterogeneity by shared demographic characteristics that we found in the analysis of breakthroughs; the experimental results suggest that heterogeneity comes from differential *responses* to exposure to a more similar peer, and not just from differential *likelihood of connection* based on demographic similarity.

Social exposure induces a positive shift in belief updates about the probability of social fit but has no effects on beliefs about admissions or graduation. Low-SES students in the treatment arm have a more optimistic distribution of belief updates about the probability of fitting in compared to low-SES students in the control arm ($p = 0.024$). We find no effects on belief updates about admissions probability or successful graduation, consistent with the finding that there are no socioeconomic discrepancies in these beliefs at baseline. Among high-SES students, the distribution of belief updates about the probability of fitting in is statistically indistinguishable between treatment and control arms ($p=0.944$). Social exposure also has no effects on high-SES students' belief updates about the likelihood of admission and graduation. Perceptions about the social environment at elite universities deter applications from low-income and first-generation students, suggesting that policies aimed at lowering admissions thresholds for disadvantaged groups may be insufficient to close socioeconomic gaps at elite universities.

Finally, we show that during one-on-one conversations with mentors, participants are more likely to seek information about student life and fitting in at university than to discuss careers, university academics, or advice about choosing a major. We surveyed mentors to

collect information about take-up (whether or not a mentor-student pair had a call) as well as the topics discussed during the calls. Mentors reported that the two most commonly discussed topics, which came up in 70% of the conversations, were “student life” and “fitting in”. In 57% of the calls, mentors indicated that “whether they would fit in and enjoy their time” was the most important factor. While descriptive in nature, these results align with our previous findings that the effects of exposure operate through changes in the perceptions of the social environment at elite universities.

Taken together, our paper shows that lack of exposure to elite universities—and the resulting negative perceptions about social fit—discourages applications from low-SES students who would see substantial labor market benefits from attending these universities. Both our results in the administrative data and our experimental results demonstrate that low-SES students’ application choices are far from immutable. Application choices are highly responsive to encountering an individual who attends an elite university. These shifts in application choices operate through learning about aspects of the university experience that are specific to a particular university and demographic group, such as the likelihood of social fit. Knowledge about whether *someone like them would fit in* at a given elite university is transmitted through direct contact with past attendees and amplified with demographic overlap. Our findings suggest that interventions offering social connectedness at scale can be an effective way to address disparities resulting from socioeconomically stratified school networks. Back-of-the-envelope calculations suggest that equalizing exposure to elite universities can close up to 30% of the socioeconomic gap in applications.

Our contributions relate most directly to the literature on the effects of peers on choices of university destinations; we provide new experimental evidence on their underlying mechanisms and new evidence on long run outcomes for students who follow their peers’ choices. Altmejd et al. (2021) find across several countries that older siblings’ university destinations affect their younger siblings’ applications and enrollment, and Barrios-Fernández (2022) and Avdeev et al. (2024) find similar effects for neighbors and other family members. Using a fuzzy regression discontinuity design in France, Bechichi and Kenedi (2024) find that an older student enrolling increases the probability of applications from that school to the same college and major by 19%, with persistent effects over time. In Norway, Cattan, Salvanes, and Tominey (2025) use idiosyncratic variation in school cohort composition (Hoxby 2000), and find that elite peer parents raise the probability of attending an elite university degree despite having a negative effect on GPA. Our results are consistent with this literature, and we build on it by providing some of the first experimental evidence on the effects of exposure, surveys and belief elicitation that isolates social fit perceptions as an important mechanism, and an analysis of long-run outcomes for students who respond

to breakthroughs.

This evidence on mechanisms additionally responds to an extensive literature documenting undermatching, wherein low-SES students attend worse universities than high-SES students with similar grades (Hoxby and Avery 2012; Black, Cortes, and Lincove 2015; Dillon and Smith 2020; Campbell et al. 2022; Barrios-Fernández 2022; Chetty, Deming, and Friedman 2026; Wyness 2023; Cattán, Salvanes, and Tominey 2025). Past literature has discussed a wide range of influences on application choices that could contribute to undermatching, including information frictions about financial aid (Dynarski et al. 2021); choice architecture (Pallais 2015); inaccurate beliefs about admissions chances (Larroucau et al. 2025); and access to school counselors and college advising (Mulhern 2023; Barr and Castleman 2025). Despite extensive work on social fit at university in psychology and sociology literatures (see for instance Jack 2020 and Walton and Cohen 2011), little work in economics has identified disparities in social fit perceptions as a mechanism for undermatching. Our unique setting and the design of our randomized controlled trial and surveys allow us to isolate this particular barrier to accessing elite universities.

We also contribute to literature evaluating interventions aimed at increasing college access and enrollment (Hoxby and Turner 2013, 2015; Andrews, Imberman, and Lovenheim 2020; Dynarski et al. 2021; Sanders, Chande, and Selley 2017; Sanders et al. 2018; Cohodes, Ho, and Robles 2022; Larroucau et al. 2025; Barr and Castleman 2025). We show that the social fit mechanisms that we discuss can be translated into scalable interventions that have substantial treatment effects on university application choices and social fit perceptions. Relative to past work that focuses on similar contact interventions (particularly Sanders, Chande, and Selley 2017 and Sanders et al. 2018), we focus not just on classes of universities as outcomes but how contact shifts applications toward a specific university, and provide further analysis of underlying mechanisms.

Our paper also relates to a broader literature on social learning beyond the economics of education, including herding models (Banerjee 1992; Eyster and Rabin 2010) and Mal-mendier, Oprea, and Veldkamp (2026) who provide a framework and experimental evidence on cases where individuals respond more to information from more similar peers. Finally, Chetty et al. (2022) show that friendships between low-SES and high-SES individuals are one of the strongest predictors of income mobility; our study provides evidence that beliefs about elite universities plays an important role in this relationship.

II. Setting and data

A. UK Educational System

The UK educational system has several features that make it uniquely advantageous to study social exposure as a barrier to university access. Further details on the UK educational system can be found in the Online Appendix [OA2](#).

First, university applications and admissions are centralized and largely based on performance on a series of standardized exams embedded in the core school curriculum. Students in England and Wales take compulsory General Certificate of Secondary Education (GCSE) exams at age 16 in mathematics, English, science, and other optional subjects.¹ Students spend the last two years of secondary education completing academic qualifications (A-levels) or vocational qualifications. Upon application, students submit a short personal statement, GCSE grades, and predicted grades in each of their A-Level or vocational subjects.² All applications go through the Universities and Colleges Applications Service (UCAS) platform. Universities independently decide whether to make an offer to an applicant or reject them outright, and final university placements are decided based in part on students' final grades.

Second, financial frictions are less relevant to university choice than they are in countries such as the US, as university tuition is uniform across universities. Tuition for domestic students was capped at £3,000 from 2006–2011 and increased to £9,000 in 2012, with irregular increases thereafter, generally below the rate of inflation. Essentially all universities charge tuition fees exactly at the cap, meaning that there is no variation in tuition between universities.³ Financial aid is also widely available and standardized, with government loans that are repaid via automatic deduction from paychecks based on income bracket.

Third, the application set is a high-stakes decision, as students can apply to *at most* five university-major pairs, called “courses” in a given cycle.⁴ This cap on the number of applications raises the likelihood that academically qualified students omit elite universities from their application portfolio.

1. We use GCSE grades in the ‘core’ subjects of maths, English and science, converted to a percentile, throughout the administrative data analysis as a measure of students’ academic ability.

2. Students typically apply to university and receive decisions before they complete their A-Level exams or vocational qualifications, so a standard component of the application system is that teachers assign their students predicted grades in each of these subjects that are reported to universities as part of students’ application. These grades are noisy and generally upward-biased (Murphy and Wyness 2020). See (Tadjfar and Vira 2025) for more details on how predicted grades and test scores differentially impact university outcomes.

3. Scottish universities have no tuition associated for Scottish students, but students from England who attend these universities pay the same tuition fees that they would pay at English universities.

4. This cap typically binds, with 80% of students in our administrative data exhausting the five courses.

Finally, school-to-university pipelines are quite dense in the UK, which restricts students' social exposure to elite universities. The Universities of Oxford and Cambridge, commonly denoted by the portmanteau "Oxbridge", enroll as many students from just 8 elite high schools as they do the remaining 3,000 (Sutton Trust 2025). Elite universities in the UK have also been widely reported to create socially isolating environments for socioeconomically disadvantaged students.⁵

B. Administrative Data

We draw on four UK administrative datasets using the Longitudinal Education Outcomes database developed by the Department for Education (Office for National Statistics 2023). The National Pupil Database (NPD) provides data on students attending English schools, including demographics, test scores at various ages, school type, and subjects taken. The Universities and Colleges Applications Service (UCAS) provides data on applications to undergraduate universities, offers, student responses to offers, and final offers accepted.⁶ For university graduation and performance outcomes, we link these with data from the Higher Education Statistics Agency (HESA), which collates annual university-level data on student enrollment, graduation, degree class, course studied, and various other details about outcomes at university. For long-run labor market outcomes, we link data from HM Revenue and Customs (HMRC). The HMRC provides data on employment spells, pseudonymized employer ID, employer industry, and annual earnings, drawn from tax records.

We focus on the sample of students completing secondary education and applying to university between 2007 and 2021, which is the timespan for which we observe all four of these datasets. Within these cohorts, we apply the following further sample restrictions. We restrict to students who apply to university as part of the UCAS "main scheme" and are aged 18 as of 31st August of the year of their application cycle and exclude students who do not apply to *any* university.⁷ Online Appendix OA1 provides additional details on variable definitions and sample restrictions.

5. See for example Bhopal and Myers (2023).

6. The UCAS dataset available in the LEO database has pseudonymised institutional identifiers, which prevents identification of particular schools or universities by name. For analysis that requires this, we draw on standalone data provided directly by UCAS that is not linkable to labor market outcomes or the NPD.

7. Specifically, we exclude students who do not apply to at least one university that can be linked with HESA data. UCAS courses that do not link with HESA data are generally more specialised courses, such as arts academies or music conservatoires, rather than traditional university courses.

C. Survey Data and Experimental Sample

We conducted a series of surveys with one cohort of UK university applicants applying in January 2026 as part of our field experiment (see Section A for details on the experimental design). We recruited 21 high schools across England and Wales to participate in our study, yielding a geographically spread out and nationally representative sample.⁸ Table 1 presents summary statistics for students participating in our experiment, as well as comparable statistics drawn from the administrative data. Compared to the national administrative data, we find there are substantial differences in ethnicity (56% of students in the RCT sample are white, compared with 80% in the general population) and gender (45% of students in the RCT are female, compared to 50% in the general population and 56% among university applicants). In terms of disadvantage, RCT participants are comparable to the overall population of age-18 school leavers and somewhat more disadvantaged than the typical university applicant. In terms of the geographical distribution, students in our RCT are more likely to be from Northern England or London, less likely to be from the Midlands, and roughly as likely to be from Southern England, reflecting the geographical distribution and size of the participating schools.

III. Socioeconomic Gaps in Elite University Enrollment

We begin our empirical analysis by documenting socioeconomic gaps in the probability of enrolling at one of the ten most academically selective universities in the UK, henceforth referred to as *elite universities*.⁹ Figure 1 reports the differences the likelihood of enrolling, applying to, and being exposed to one of the elite universities, between low-income and high-income students. We focus on students above the 90th percentile of age-16 standardized test scores (GCSE grades) since application and admission rates at elite universities are very low below this level.

The highest-achieving students from low-income backgrounds are up to 12 percentage points less likely to enroll at an elite university compared to high-income students with the same grades. Crucially, this disparity largely begins at the application stage, as low-

8. Schools were recruited to participate in the programme through various channels, including school networks from our partner organisation, WISE/IET, as well as direct outreach to schools. WISE (Women Into Science and Engineering) is a UK social enterprise that aims to support women’s access to STEM university courses and careers, but has a network of coeducational schools that we worked with to recruit students. We collected data through Qualtrics surveys administered to students at home and in school, as well as data on student applications and enrollment directly from participating schools.

9. Universities are ranked using the mean A-level grades of students enrolled at each university as a measure of the university’s selectivity.

income students are up to 12 percentage points less likely to *apply* to an elite university. If low-income students applied to elite universities at the same rate as high-income students, holding offer rates and enrollment conditional on receiving an offer fixed, enrollment disparities for the highest-ability students would fall by 40-50%.¹⁰ At the same time, low-income students are substantially less likely to encounter individuals attend elite universities, both through family members and also through schoolmates. High-achieving low-income students are 6 percentage points *less likely* to attend a school where a student has attended an elite university sometime in the last three years.

IV. A Simple Framework of Social Exposure and University Choice

We provide a simple framework to illustrate how systematic lack of exposure to elite universities can deter applications among future cohorts within a school.

A. Setup

Time is discrete and indexed by cohorts $t = 1, 2, \dots$; for simplicity we assume there is one student per cohort choosing between an elite university and a local default. The payoff to student t from attending an elite university is

$$(1) \quad v_t = \theta + \gamma + e_t,$$

where θ is a publicly known common return, $\gamma \sim N(0, \sigma_\gamma^2)$ is an unobservable *community-level match component*—how students from this community fare at the university—and $e_t \sim N(0, \sigma_e^2)$, i.i.d. across cohorts, is an *individual component* that student t observes privately before choosing. The student’s outside option is a known local default with value $\bar{u}$, so students choose by comparing the unknown components of payoff $\gamma + e_t$ to $\tau \equiv \bar{u} - \theta$.

Students have CARA expected utility preferences with risk aversion λ . Given a posterior distribution of $\gamma \sim N(\mu_t, \sigma_t^2)$, a student attends the elite university if and only if

$$(2) \quad e_t \geq \kappa_t \equiv \tau - \mu_t + \frac{\lambda}{2} \sigma_t^2.$$

10. The probability of enrolling at a given university through the main UCAS scheme can be decomposed as $P(\text{enroll}) = P(\text{apply}) \times P(\text{offer} \mid \text{apply}) \times P(\text{accept} \mid \text{offer}) \times P(\text{enroll} \mid \text{accept})$. Holding the last three terms at their low-income values and replacing $P(\text{apply})$ with the high-income application rate gives the counterfactual enrollment rates quoted here; taking logs gives an additive decomposition in which application gaps account for roughly 40–50% of the enrollment gap.

B. Signals From Peer Choices

Student t observes the previous cohort's action $a_{t-1} \in \{0, 1\}$. If the predecessor attended, her realized payoff is also observed, revealing $x = \gamma + e_{t-1}$. If she did not attend, nothing further is observed.

Each cohort updates from the common prior using a simple theory of the predecessor's behavior: it treats the predecessor as having known her own total match $\gamma + e_{t-1}$ and attended if and only if it exceeded τ . Cohorts model their predecessors as making decisions under certainty, rather than as uncertain social learners themselves (this naïve interpretation of predecessors' actions follows Eyster and Rabin (2010)). Without this assumption, a history of non-attendance would carry no information about γ if social learners were themselves uncertain. Let $\sigma^2 \equiv \sigma_\gamma^2 + \sigma_e^2$, $r \equiv \sigma_\gamma^2 / \sigma^2$, $\beta \equiv \tau / \sigma$, and let $h(\beta) \equiv \phi(\beta) / \Phi(\beta)$ denote the inverse Mills ratio.

Observing non-attendance. A student who observes $a_{t-1} = 0$ infers (possibly incorrectly) that $\{\gamma + e_{t-1} < \tau\}$. The event truncates beliefs about $\gamma + e_{t-1}$; projecting onto γ then yields¹¹

$$(3) \quad \mu^- = -r\sigma h(\beta) < 0, \quad (\sigma^-)^2 = \sigma_\gamma^2 [1 - r(\beta h(\beta) + h(\beta)^2)].$$

so non-attendance is an unambiguously negative signal. As the payoff is not observed directly, the variance reduction is strictly smaller than that from observing an attendee (below); under risk aversion, this is an additional force that leads non-attendance to discourage future applications.

Observing attendance. The realized payoff reveals $x = \gamma + e_{t-1}$ exactly. The observer cannot decompose the sum, and optimally attributes fraction r of it to the community:

$$(4) \quad \mu^+ = rx, \quad (\sigma^+)^2 = \sigma_\gamma^2 (1 - r) < (\sigma^-)^2.$$

Attendance may move the mean in either direction depending on the realized outcome, but always delivers the larger variance reduction.¹²

11. The posterior on γ itself is not truncated — any value of γ remains possible if the predecessor's individual draw was sufficiently low — but is smoothly shifted left, with density proportional to $\phi(\gamma/\sigma_\gamma) \Phi((\tau - \gamma)/\sigma_e)$. The moments in (3) are exact.

12. We can relax the assumption that sharing a community is a binary by introducing imperfect correlation between community payoffs: assume γ_{t-1} and γ_t are jointly normal with variance σ_γ^2 and covariance $\rho \in [0, 1]$. A factor structure generates ρ from primitives: let $\gamma_i = s'_i g$, where g collects match components shared at different levels of social aggregation (school, neighbourhood, class background, gender, ethnicity),

C. Dynamics: Persistent Non-Attendance and Breakthrough Students

Because each cohort starts from the prior and observes only the adjacent cohort, behavior follows a two-regime Markov process with cutoffs

$$\kappa^- \equiv \tau + \frac{\lambda}{2}\sigma_\gamma^2 + \lambda^{-1} \log \left[\frac{\Phi(\beta + \lambda r \sigma)}{\Phi(\beta)} \right], \quad \kappa^+(x) \equiv \tau - \mu^+ + \frac{\lambda}{2}(\sigma^+)^2,$$

implied by (2)–(4). Let $\kappa_0 \equiv \tau + \frac{\lambda}{2}\sigma_\gamma^2$ denote the cutoff under prior beliefs.

Proposition 1 (Persistence of non-attendance). $\kappa^- > \kappa_0$: following a non-attendance, the probability of a further non-attendance strictly exceeds the probability of non-attendance under prior beliefs.

Non-attendance in period $t - 1$ thus increases non-attendance in period t , and can originate in a single unlucky individual draw: an idiosyncratic reason not to attend is misread by the next cohort as evidence about the community match, raising its own bar to attend, and so on. Conditional on true γ , beliefs in the non-attendance regime are systematically too pessimistic (as long as true γ is weakly positive) and leads to herding where communities with the same γ make persistently different choices based on early idiosyncratic history.

Proposition 2 (Breakthrough). Upon attendance with realized outcome $x = \gamma + e_{t-1}$, the next cohort's beliefs jump to (4): the mean is re-anchored to $\mu^+ = rx$, a quantity containing the true γ , and the variance falls discretely, $(\sigma^+)^2 < (\sigma^-)^2$ and following any breakthrough with outcome above $\bar{x} \equiv -\sigma h(\beta) - \frac{\lambda}{2r}[(\sigma^-)^2 - (\sigma^+)^2] < 0$, the probability of attendance strictly exceeds its level in the non-attendance regime.¹³

We refer to a student who exits the non-attendance regime as a *breakthrough student*; these individuals have sufficiently high idiosyncratic draws to overcome their community's pessimism.

$g \sim N(0, \Sigma)$, and loadings are normalised so that $s_i' \Sigma s_i = \sigma_\gamma^2$ for all i . Then $\rho_{ij} = s_i' \Sigma s_j / \sigma_\gamma^2$: similarity is overlap in group membership. Weighting others' experience by the covariance between their circumstances and one's own parallels the technology-learning weights in Conley and Udry (2010). For $\rho < 1$ the observer extracts information about γ_i from $\gamma_j + e_j$. Then $\mu^- = -\rho r \sigma h(\beta)$, $(\sigma^-)^2 = \sigma_\gamma^2 [1 - \rho^2 r (\beta h(\beta) + h(\beta)^2)]$, $\mu^+ = \rho r x$, $(\sigma^+)^2 = \sigma_\gamma^2 (1 - \rho^2 r)$ with $(\sigma^+)^2 < (\sigma^-)^2$ for every $\rho > 0$.

13. The non-attendance regime is persistent but not absorbing. In each cohort of the regime, attendance occurs with probability $1 - \Phi(\kappa^- / \sigma_e) \in (0, 1)$, so the regime is exited in finite time almost surely.

V. Breakthroughs to Elite Universities

A. Empirical Specification

To understand whether and how social exposure to elite universities influence application choices, we use quasi-random variation in exposure generated by university-specific “breakthrough” events schools. Consider two schools, School A and School B, that have both had no student attend University X for several years. If a student from School A attends University X, we refer to this as a breakthrough event to University X at School A, and refer to the student who first attends as a breakthrough student. This is a discrete change in the exposure to University X for future cohorts at School A. We examine how university applications at school A change relative to application patterns at school B following this breakthrough.

We pool our analysis in an event study design to exploit this variation across the large number of schools and universities in UK administrative data. To implement the strategy outlined above, we identify, for each university, the schools where no students enrolled at the university between 2007 and 2009, the first three years of our data. We then define the first year a student attends that university from that school as the breakthrough year for that school. If no student attends the university during the period covered in our data – from 2010 to 2021 – we assign that school to the control group for that university. By construction, no students from any school in the sample are enrolled at that university before their school’s event year, and no students from control schools are enrolled at that university at any point in the sample. Our estimating equation is then:

$$(5) \quad Y_{ist} = \alpha_s + \gamma_t + \delta X_{ist} + \sum_{\tau \neq -1} \beta_\tau \mathbb{I}(t - T_s = \tau) D_s + \varepsilon_{ist}$$

where X_{ist} is a vector of individual-specific covariates, T_s is the year in which school s had a breakthrough and D_s is the treatment indicator. Our primary specification omits individual-level covariates to show trends in applications at the affected universities transparently, but in robustness checks included in appendix figures, we include in X_{ist} students’ GCSE percentile (see Appendix OA1) and indicators for the number of A-levels they took, as well as the number of A-levels in facilitating subjects.¹⁴ We exclude the breakthrough student from the sample (or if there are multiple breakthroughs to the same university in the same year, we drop one of the breakthrough students), in order to isolate the effect of a breakthrough on the rest of the breakthrough student’s cohort in period $\tau = 0$. To allow

14. In 2011, the Russell Group of universities published a list of “facilitating subjects” important for selective universities.

for heterogeneous treatment effects by treatment year, we use the Sun and Abraham (2021) estimator for event studies.

To provide a more aggregated picture of the effects of breakthroughs, we stack breakthroughs across 10 elite universities. We construct a dataset for each university and stack them, indexing data from each by the breakthrough university u , and then run the following stacked event study regression¹⁵:

$$(6) \quad Y_{istu} = \alpha_{su} + \gamma_{tu} + \delta_u X_{istu} + \sum_{\tau \neq -1} \beta_{\tau} \mathbb{I}(t - T_{su} = \tau) D_{su} + \varepsilon_{istu}$$

We also report results for certain outcomes using the analogous difference-in-differences specification of these effects, which pools effects across the post-treatment periods:

$$(7) \quad Y_{istu} = \alpha_{su} + \gamma_{tu} + \delta_u X_{istu} + \beta \mathbb{I}(t - T_{su} \geq 0) D_{su} + \varepsilon_{istu}$$

B. Shifts in Applications of Future Cohorts

First, we present an illustrative example using breakthroughs to the universities of Oxford and Cambridge. We use these two universities as an example of the university-level event study design for two reasons: first, they are generally considered to be particularly elite universities; second, they are generally seen as *equally* selective and elite, to the extent that they are often just referred to by the portmanteau ‘Oxbridge’. We see from this analysis that a breakthrough to Cambridge encourages applications to Cambridge, but not to Oxford, and vice versa. Figure 2a plots coefficients on the relative time indicators from (5), where the treated schools are those that experience a breakthrough to Cambridge and the outcomes are application to Oxford and Cambridge as indicated; Figure 2b does the same, but for schools that experience a breakthrough to Oxford. We see an increase in applications to Cambridge in panel (a) at the year of the breakthrough to Cambridge, rising by 0.6–0.8 percentage points, while the application rate to Oxford does not significantly increase. Similarly, in panel (b), there is an increase of 0.4–0.8pp in applications to Oxford following the breakthrough to Oxford while the application rate to Cambridge stays largely constant. The effects persist at least four years after the breakthrough, which may reflect

15. A given student may appear multiple times in the stacked dataset if their school sees breakthroughs to, or is in the control group for, multiple universities. We cluster standard errors at the school level, which accounts for correlation within a school across the breakthrough university samples (Wing, Freedman, and Hollingsworth 2024).

the creation of a pipeline, in which students apply to and attend the university at higher rates in the years after the breakthrough.

This pattern illustrates our key finding, which generalizes across universities: following a breakthrough to a particular university at a school, applications to that university persistently increase, but applications to comparable, similarly-ranked universities do not. This rules out any explanation that attributes the increase in applications following a breakthrough to a common shock that would affect applications to a range of universities. *Ex ante*, many plausible explanations for the effect of breakthroughs to Cambridge on applications to Cambridge would center on information about these procedures that would also be informative about and encourage applications to Oxford. Yet instead we find effects concentrated at Cambridge, and vice versa for breakthroughs to Oxford; any mechanism that explains the effects of breakthroughs must therefore be highly *university-specific*.

If applications to breakthrough universities increase following a breakthrough, this must draw applications away from other universities, given that the cap of five applications is typically binding in the UK. To understand where applications are drawn from, we construct a set of mutually exclusive and exhaustive outcomes based on students' portfolios of five applications. These are portfolios that include the breakthrough university; portfolios that exclude the breakthrough university, but include a different university in the same 5-university tier; portfolios that include no universities from the breakthrough tier but include at least one university from a higher tier; and portfolios that include only applications from lower tiers than the breakthrough university.

Figure 2c reports the results of this analysis, pooling across the top 10 universities. First, Figure 2c shows that the breakthrough effects to Cambridge and Oxford generalize to a broader set of elite universities. We see substantial and persistent increase in applications of 0.4–0.6 percentage points pooled across these universities. We also observe that there is no change in the probability of applications to higher-ranked universities or universities of the same rank, consistent with the result that applications to Oxford do not change in response to a breakthrough to Cambridge and vice versa. Instead, the applications to the breakthrough draw applications away from lower-ranked universities. Breakthroughs to elite universities increase the ambition of future students' application portfolios. Students exposed to a breakthrough event at their school apply to the breakthrough university when they would otherwise have applied to lower-ranked universities.

To benchmark the size of these effects, we quantify the effect that equalizing exposure across income groups would have on application rates given the observed effect sizes using a back-of-the-envelope exercise. We find that around 30% of the overall socioeconomic disparity in applications can be explained by differences in exposure, with the largest effects

among students with the highest GCSE grades. Details of the calculation are provided in Online Appendix Section [OA4](#).

Finally, we assess the extent to which an increase in applications goes on to increase enrollment at and graduation from the university in question. Figure [2d](#) reports event study coefficients for applications, receiving an admission offer, enrolling, and graduating successfully from the breakthrough university. Enrollment increases by 0.3–0.4 percentage points and constitutes 60–85% of the increase in applications. Furthermore, among those induced to enrol by the breakthrough, almost all students graduate successfully, as the coefficients in the enrolment and graduation event studies are nearly identical.

C. Early-career Earnings Outcomes

Virtually all of the students who enroll at an elite university following a breakthrough go on to graduate successfully. These students may nevertheless fail to capture labor market gains from attending an elite university (for instance, see Michelman, Price, and Zimmerman [2022](#)). To address these concerns, we use a matched controls design to directly examine earnings gains several years out of university.¹⁶

We identify students who apply to the breakthrough university following a breakthrough at their school and compare them with untreated students who are matched on a wide set of covariates.¹⁷ We conduct two different matching procedures. In the first, induced applicants are matched with students from untreated schools applying to university for the same major in the same year; in the second, induced applicants are matched with students from the same school in the pre-breakthrough cohorts. Within each of these sets, we match exactly on students' quintile of neighborhood income, ventile of standardized test scores (GCSE grades), and an indicator for taking at least three A-Levels. Finally, we select one nearest neighbor from within the exactly matched set for each student, based on the Mahalanobis distance over gender, ethnicity, region, and standardized test scores (GCSE grades). Induced applicants who have no available exact matches on the relevant variables are discarded from the estimation.

With these matched control and treatment observations, we estimate the returns to

16. Given the relatively small magnitude of breakthrough effects in absolute terms, event studies using earnings as the outcome are underpowered. Similarly, a 2SLS regression instrumenting for applying to the breakthrough university with the post-breakthrough indicator yields a 95% confidence interval of around (−£12000, +£92000) for effects on age-27 earnings, which includes implausibly high positive and negative effects.

17. We refer to these students as 'induced applicants', though note that this set of students includes both compliers (who apply to the breakthrough university only because of the breakthrough at their school) and always-takers (who would have applied even without the breakthrough). Note that this includes all induced applicants, so we are not restricting to students who successfully place at the breakthrough university.

applying to an elite university by regressing earnings at ages 22, 25, and 27 on matched-pair fixed effects and an indicator for applying. We report estimates in Figure 3. By age 27, induced applicants have earnings £4,414 higher than matched controls from control group schools, and £4,003 higher than matched controls from pre-treatment years at their own school.¹⁸ A simple back-of-the-envelope calculation yields a discounted lifetime earnings effect of £92,362,¹⁹ which substantially exceed tuition fees.

D. Heterogeneity by Demographic Similarity and Social Proximity

How do breakthrough effects vary by extent of exposure and demographic similarity with the breakthrough student? We first consider demographic overlap in low-income status, free school meal (FSM) eligibility, gender, and ethnicity separately. Next, we examine heterogeneity by exposure using two proxies for exposure intensity: an indicator for attending same school as the breakthrough student prior to age 16 and the number of overlapping subjects with the breakthrough student.²⁰ For each characteristic, we estimate a difference-in-differences version of the breakthrough design and interact the main difference-in-differences indicator with a binary indicator for sharing the specified characteristic with the breakthrough student. Figure 4 reports estimates of the main effect and the interaction effect for each dimension for breakthroughs elite universities.

We find that both demographic overlap and extent of exposure matter for breakthrough effect sizes. Demographic overlap raises the effect of a breakthrough by around 0.1–0.25 percentage points (around 30–50% of the total effect), and stronger exposure raises the effect of a breakthrough by 0.5–0.75 percentage points (64–80% of the total effect). These patterns are consistent with the transmission of information about some aspect of the university experience or application process through conversations with peers. Our results on demographic similarity may be capturing the fact that there is demographic homophily in the likelihood

18. As a robustness exercise, we estimate effect sizes for various universities using an empirical strategy similar to Dale and Krueger (2002) controlling for application portfolios. We report these Appendix Figure OA5c. We find that the average graduate of a top 10 university earns £4,600 more than a student attending a university near the median of the quality ranking, which is broadly consistent with the matched-control estimates.

19. We take the point estimates of earnings premia using each approach at ages 22, 25, and 27, and linearly interpolate earnings premia between these ages. After age 27, we assume there is no real growth in the earnings premium for induced applicants, so this premium remains constant until retirement at age 68; this is a conservative assumption given the substantial increase in the earnings premium between even ages 25 and 27. We discount the earnings premium at 3% per year back to age 22, following Angrist, Autor, and Pallais (2022).

20. We focus on school at age 16 because it is common—but not ubiquitous—for students in the UK to change school after completing their GCSEs at age 16, and attend a different school for the final two years of their education.

of friendship formation. Alternatively, larger effect sizes among demographically similar peers may reflect learning about group-specific university payoffs.

E. Validity

Common School-level Shocks.— One potential confounder that threatens our interpretation of the breakthrough effects is the presence of common school-level shocks. For instance, schools may receive sudden boosts in funding, improvement in student quality, or receive an influx of better teachers or guidance counselors. In Figure 2, breakthroughs induce an increase in applications to the specific university that sees a breakthrough, but not to other equally selective universities. This pattern rules out common shocks such as a sudden increase in students’ ability or academic performance, as such shocks should lead to a broader increase in the selectivity of applications. Furthermore, Table OA1 shows that our results are robust to controlling for various measures of academic ability.²¹

University-specific School-level Shocks.— Another more challenging confounder is a *university-specific* common shock affecting all students at the school. For example, a breakthrough may coincide with the year when a university does an outreach event at that high school. Outreach activities are generally targeted within the local area of the university.²² If breakthrough effects were driven by outreach, the effects would be concentrated in the geographical area of the university. Table OA2 indicates that this is not the case. Yet another possibility is that breakthroughs are caused by the arrival of a new teacher who has a connection to a specific university and encourages applications to this university. Figure 4 shows that there is a statistically significant effect of breakthroughs even on students who had no subjects in common with the breakthrough student and who were thus unlikely to interact with the same teacher. Ultimately, however, fundamental limitations of the administrative data mean that it is challenging to entirely rule out alternative mechanisms. Our field experiment addresses this limitation by inducing precisely defined exposure to past university attendees and measuring effects on beliefs and application choices.

21. We additionally report coefficients from a matched event study design. In the matched specification, we conduct 1:1 matching of treatment and control schools based on students’ university application behaviour and ability in the pre-breakthrough window, 2007–2009, and include match-pair by year fixed effects in the specification, so effects are identified from changes in outcomes for treated schools relative to a matched control school. This reduces pre-treatment disparities in outcomes between treatment and control schools.

22. University outreach to schools is often coordinated through the ‘Uni Connect’ Programme, which is organised around 29 regional hubs across England. This programme connects universities and schools within geographical regions of the UK but does less to coordinate outreach across regions. See Burtonshaw et al. 2024 for an evaluation of Uni Connect.

VI. Evidence from the Field

Which information frictions does social exposure alleviate to shift university application decisions? Our field experiment is designed to isolate potential mechanisms such as changes in beliefs about admissions chances, social fit and sense of belonging, and returns to university and rule out potential confounds from our findings in the administrative data.

A. Experimental Design

We recruited 21 high schools in the UK to participate in our field experiment, which randomizes exposure to current university students at academically selective universities. Panel A in Figure 5 overlays our participating schools on a map of England shaded based on the local probability of not being exposed to a top 10 university.²³ The experiment consisted of a baseline survey (Jan–Jun 2025), an in-person workshop led by a school alumnus or current university student and a midline survey (Apr–Jun 2025), an endline survey (Oct 2025), and realized university applications data collected directly from schools (Mar–Oct 2026).²⁴ Participants randomized into one of the treatment arms additionally watch video clips from current university students and receive one-on-one mentorship through calls and emails with university students.²⁵ Panel B in Figure 5 provides an overview of the experimental design. Randomization was at the individual-level and participants were assigned to treatments at the beginning of the baseline survey. Table OA6 presents balance checks among our primary sample of students who completed both the baseline and midline surveys, pooling across the treatment arms; all covariates that we test are balanced across treatment and control groups. Our main analysis sample consists of 640 participants who were in their final year of study (Year 13), expressed some interest in applying to universities in January 2026, and took both the baseline survey and midline survey.²⁶ Appendix A provides additional details on the experiment.

Active Control: In-person Workshops— We invited all participants, regardless of treatment assignment, to attend a one-hour in-person workshop delivered by a current university

23. We have one participating school in Wales which is omitted from this map, since our LEO data on university access only covers England.

24. As of September 9, 2026, we have collected final outcomes for 473 of the 640 students. Some schools are still finalizing the data collection process.

25. Note that in our pre-registration, we also described a treatment offering subsidised in-person visits to universities. For various logistical reasons, this arm was substantially underpowered and we do not report results from this arm.

26. Students who indicated they will not apply to university terminated the baseline survey early and were not part of our study.

student held at their school. This workshop was designed to ensure that participants had a baseline level of information about the university application process and provided statistics on average earnings of graduates from different universities and typical admissions requirements at various universities.²⁷ In principle, this helps rule out that any treatment effects estimated from the experiment are explained by changes in beliefs about the average returns to university or admissions criteria as this information was provided to all participants. The workshop also discussed the application process and timeline; student loans, cost of living and student finance; and how to write personal statements. The full set of workshop slides can be found in Online Appendix [OA9](#).

Treatment: Videos from University Students.—This treatment is designed to replicate, at scale, the information that a student could learn about a particular university from a conversation with a schoolmate who went to that university. At the end of the baseline survey, participants randomized into treatment watch two video clips recorded by a subset of our mentors. We group participants into one of three academic tiers based on their grades and assign them videos within their academic tier. This ensures that treated participants watch videos from universities that are aspirational but not academically out of reach given their grades and coursework. Table [OA7](#) presents the full set of video universities and the grades corresponding to each tier. Each video is 3-4 minutes long and discusses the speaker’s university application process and experience at their particular university.²⁸

Treatment: Demographically Matched vs. Unmatched Mentors.—This treatment arm is designed to test whether participants respond differently to information from a demographically *dissimilar* university student compared to a demographically *similar* university

27. In addition to calibrating student beliefs by providing statistics about earnings and grade requirements for admissions, the workshops also served several logistical functions. Having a component of the program available to all students rather than just the treatment group was important for school recruitment, as schools would have been less enthusiastic about a program in which a large fraction of their students would get no benefit. Collect midline survey responses in person during the workshop and communicating mentor assignments also helped reduce attrition.

28. Specifically, mentors were told to answer the following: “Say your first name, secondary school attended, and city you’re from”, “Which A-level / BTEC subjects did you take? If comfortable sharing, predicted and achieved grades in A-levels / BTECs and how those affected your choices”, “Subjects you applied to, universities you applied to, where you got offers from, where you chose as your firm / insurance choice and why?”, “Who you talked about the university application process with / resources you consulted”, “How you approached your personal statement”, “Anything that surprised you about the process?”, “Do you enjoy the city and student life in the area?”, “Do you have a good sense of community at your uni?”, “Have you found it easy to make friends at uni?”, “What is your living situation, approximately how much does it cost per month?”, “Do you feel it is financially affordable?”, “How is the teaching quality and academic experience at your uni? Do you feel prepared to handle the rigour?”, “Did you know many other people with similar backgrounds going to your university?”

student. Students randomized into the treatment arm T at baseline are further randomized, with equal probability at the individual level, into one of T1 and T2. Students in T1 are assigned up to two mentors that are demographically dissimilar, i.e., mentors who overlap in *none* of the following characteristics: gender, region of hometown, and ethnicity. Students in T2 are assigned up to two mentors that are demographically similar, i.e., mentors who overlap in *at least one* of the following characteristics: gender, region of hometown, ethnicity.²⁹ Participants in T1 and T2 receive their mentor assignments at the end of the in-person workshop after completing the midline survey. Mentors reach out to students over email and schedule time for a call. We recruited 245 volunteer mentors who either currently attend or previously attended a UK university for their undergraduate degree.³⁰ Table OA8 presents summary statistics for our mentor pool; the vast majority of our mentors are current university students under the age of 25.

Outcomes.— The baseline, midline, and endline surveys collect the universities a student wishes to apply to, as well as their beliefs about various academic and social aspects of universities, allowing us to document belief evolution and changes in application portfolios over the course of the experiment. Specifically, for each of Oxford University and one of the video universities (randomly selected), the surveys ask students to report the percentage chance that they would (a) get an offer from, (b) make friends and fit in at, or (c) graduate successfully from each university. Student demographic characteristics and academic history (e.g., teacher-predicted grades and coursework) are collected in the baseline and midline surveys. We collect realized application outcomes directly from schools.

B. Socioeconomic Gaps in Baseline Beliefs about Admissions, Academic Success, and Social Fit

We begin by exploring socioeconomic gaps in beliefs about Oxford University as a benchmark for elite universities in our experimental sample.³¹ Panel A in Figure 6 shows that low-SES students, defined here as either first-generation university attendees or low-

29. Participants listed up to three universities from which they would like to receive a mentor. Mentor assignment was subject to capacity: one from one of the universities that they requested, and one from an academically matched university that they did *not* request, where we use the same procedure to academically match participating students and universities as we do for the video treatment. Appendix A provides additional details on our mentor assignment algorithm.

30. Mentors were recruited from a variety of sources, including Sutton Trust, directly from several universities, DiscoverEconomics, as well as from STEM Ambassadors, a STEM-focused volunteering platform that we partnered with for recruitment.

31. We ask about Oxford to have a benchmark for students' beliefs about a particular elite university that all students are asked about; none of our participating schools are in or near Oxford, so no schools would have a particular local connection to this university.

income students, have pessimistic beliefs about their probability of fitting in at Oxford compared to their high-SES counterparts with similar grades (we control for A-level tariff points in the regressions). On average, low-SES students have a 9 percentage point ($p = 0.001$) lower perceived probability of fitting in and making friends at Oxford than other students. However, we do not see similar patterns for the probability of receiving an offer or graduating successfully from Oxford.

Our baseline surveys also ask students to report what they believe are “typical” A-level grades achieved by students who attend Oxford University. This allows us to compare students’ perceptions of academic requirements against ground truth measures of admitted cohorts. Panel B in Figure 6 shows that low-SES students are well-calibrated about grade requirements at Oxford. If anything, low-SES students are slightly overly optimistic, with around 30% estimating grades that are below the 25th percentile.

These patterns are consistent with the notion that lack of exposure in the sense of persistently observing *non-attendance* to an elite university can generate pessimism as outlined in our framework in Section IV. Moreover, this lack of exposure primarily affects beliefs about aspects of universities that are demographic-group specific and highly university specific, such as social fit. We do not find any evidence that less exposed students are pessimistic or otherwise less informed about more general aspects of elite universities that are likely to be common across demographic groups (and transferable across universities).

C. Results: Shifts in University Applications

Our primary outcomes are indicators for whether a student applied to (i) either of their two assigned video universities, (ii) either of their assigned mentor universities, and (iii) any of these assigned “treatment” universities. We assign latent video and mentor universities to all participants to ensure that outcomes are defined regardless of treatment status. We measure applications using records that we collect directly from schools, supplemented with students’ endline reports of their submitted applications where school records are not yet available.³² Our preferred specifications report estimates among the 567 students (89% of the sample) who did not explicitly indicate that they do not wish to participate in the mentorship treatment; results are similar, but slightly noisier, when estimated on the full sample (Appendix Figures OA2 and OA3).³³

32. Of the 473 students in our sample with endline outcomes, 395 are sourced from school records and 78 are supplemented with endline survey results. Our results are very similar when we exclude the 78 students.

33. Specifically, we exclude students who answered “Definitely not” or “Probably not” when asked at baseline whether they would like to be matched with a mentor. Estimates in the full sample ($N = 640$) are positive but attenuated, consistent with limited scope for effects among students with no interest in engaging with the program.

Intent-to-Treat.—We run the following intent-to-treat (ITT) specification:

$$(8) \quad Y_i = \alpha + \beta T_i + \varepsilon_i$$

where Y_i denotes the application outcome of interest for student i and T_i is an indicator for being randomized into either treatment arm (pooling T1 and T2).³⁴ Since T_i is randomly assigned, β identifies the causal effect of *assignment* to videos and mentors regardless of whether the student took up the treatment. When we estimate effects separately for the two mentor arms, we replace T_i with separate indicators for assignment to T1 (demographically dissimilar mentors) and T2 (demographically matched mentors).

Panel (a) of Figure 7 presents the ITT estimates. Treatment assignment increases the probability of applying to at least one assigned treatment university by 14 percentage points ($p = 0.004$). Turning to the individual components, treated students are 10 percentage points ($p = 0.008$) more likely to apply to one of their two assigned video universities and 11 percentage points ($p = 0.032$) more likely to apply to one of the mentor universities. The result that both videos and mentors are about equally as effective in inducing applications suggests that the breakthroughs discussed in Section A are not driven by students wishing to attend a university at which they already have a friend. While the mentor treatment could plausibly generate a friendship, watching a video clip without interacting with the speaker does not. This indicates that it is *new information* rather than a new friendship that generates shifts in university applications.

Assignment to a demographically matched mentor (T2) increases applications to the mentor universities by 21 percentage points ($p = 0.001$), whereas assignment to a demographically dissimilar mentor (T1) has no detectable effect, with a point estimate of 0.4 percentage points ($p = 0.952$). Our breakthrough results (Section A) could not distinguish between demographic *homophily in friendship* (i.e., students are more likely to interact with the breakthrough student when demographically similar) and *differential responsiveness* to information as an underlying mechanism for the observed heterogeneity by demographic overlap. Our experimental results reveal that students are indeed more responsive to information when it comes from a demographically similar mentor.

Local Average Treatment Effects.—We instrument take-up with random assignment and

34. We pre-registered relevant pooled comparisons between these arms, as well as the comparisons of individual treatment arms. Following the discussion in Muralidharan, Romero, and Wüthrich (2025), pooled comparisons should be interpreted as a weighted average of the effects of one treatment averaging over assignment to other treatments – so the pooled comparison of (T1, T2) vs. C can be interpreted as the effects of mentor access pooling across whether or not the student was assigned a demographically similar mentor.

estimate the local average treatment effect using two-stage least squares (2SLS):

$$(9) \quad D_i = \gamma_0 + \gamma_1 T_i + u_i$$

$$(10) \quad Y_i = \delta_0 + \delta \widehat{D}_i + v_i$$

where D_i denotes take-up of the treatment corresponding to the outcome (watching the assigned videos for the video-university outcome, any interaction with a mentor for the mentor-university outcome, and either form of engagement for the pooled outcome).³⁵ The coefficient δ identifies the local average treatment effect (LATE) for compliers (Imbens and Angrist 1994); with one-sided noncompliance, compliers are simply treated students who take up the treatment, so the LATE coincides with the average effect of treatment on the treated (TOT). For the arm-specific mentor estimates, we restrict the sample to the control group and the relevant arm and instrument take-up with that arm’s assignment indicator.

Panel (b) of Figure 7 presents the LATE estimates. Students who take up either treatment are 21-22 percentage points more likely to apply to a treatment university. Interacting with a demographically matched mentor raises the probability of applying to a mentor university by 49 percentage points ($p = 0.001$) among the treated, while the corresponding estimate for demographically dissimilar mentors is null. Take-up was higher in the dissimilar arm (42%) than in the similar arm (35%), suggesting that the null effect of dissimilar mentors is not driven by differences in take-up.

D. Results: Heterogeneity by SES

Our framework predicts that treatment effects should be concentrated among low-SES students: they are the students who lack social exposure to elite universities and, as documented in Section B, the students who hold pessimistic beliefs about their social fit at such universities. The video and mentor treatments provide precisely the university-specific social exposure that low-SES students may lack in their existing networks. Figure 8 presents ITT and TOT estimates separately by socioeconomic status. We find that treatment effects are concentrated among low-SES students across both video and mentor treatments. Low-SES point estimates are two to four times larger than high-SES ones across outcomes. In the control group, low-SES students apply to their assigned video universities at a rate of 11%, compared with 17% among high-SES students; the treatment effect for low-SES students is large enough to close this baseline SES gap in applications.

35. We designate students who spend at least 100 seconds on each video as having “watched” the videos. Results are robust to longer horizons such as 200 seconds.

E. Mechanisms: Shifts in Beliefs about Social Fit

The socioeconomic gaps in social beliefs about Oxford at baseline along with large treatment effects among low-SES students suggest that changes in application behavior are likely mediated by rightward shifts in beliefs about the social environment at elite universities.³⁶ We now directly examine treatment effects on beliefs about admissions chances, social fit, and graduation probability separately by SES.

Figure 9 illustrates the full distributions of belief updates between the baseline and midline survey for low-SES students. We see a distinct rightward shift in the distribution of belief updates about fitting in and making friends for students who received the video treatment relative to those who didn't, shown in panel (a), with no effect on the probability of receiving an offer or graduating successfully (panels (b) and (c)). We reject equality in distributions between treatment and control for social belief updates ($p = 0.024$), but not for beliefs about offer likelihood ($p = 0.732$) or graduation probability ($p = 0.386$). We find no effects among high-SES students in any of the three beliefs (Figure 10). A placebo analysis finds no statistically significant shifts in any of the three beliefs about Oxford University among low-SES students (Figure OA1).³⁷ As a robustness check, Table OA5 reports regression estimates with positive updates, negative updates, no updates, and belief gaps as outcomes. We find no statistically significant treatment effects on any outcome among high-SES students. Among low-SES students, we find positive treatment effects on positive updates and negative treatment effects on negative updates.

F. Mechanisms: Topics Discussed with Mentors

In addition to these outcomes collected from student surveys, we conduct surveys of mentors who have contacted students, and ask them to provide information about their discussions with each of their mentees. These outcomes are only available in the treatment groups where students were matched with mentors, so we cannot observe treatment effects on these outcomes. We instead use them to provide descriptive evidence of the topics that students are interested in discussing when interacting with past university attendees.³⁸

36. If lack of exposure resulted in uncertainty, university application choices may still shift in response to acquisition of new information among risk averse students.

37. None of the videos discussed Oxford, so university-specific belief updating should have no effect on beliefs about Oxford. However, similarly elite universities (including the University of Cambridge) were included, so the fact that we find no updating about Oxford indicates the university-specific nature of belief shifts, consistent with our previous findings with breakthrough students in Section A.

38. As discussed in section A, we did not prescribe topics for students to talk about with mentors. The topics that students discuss with their mentors thus reflect what students are most interested in learning from past university attendees and choose to discuss with them. They are incentivised in the sense that

Table 2 reports the topics that students discuss with mentors, as reported by mentors in a survey.³⁹ The two most commonly discussed topics are student life at the mentor’s university (70% of conversations) and life at university and fitting in (68% of conversations). Thus in these organic conversations students are most likely to want to discuss aspects of social fit, substantially more than advice about choosing applications (53%), careers after university (36%), or how to succeed academically at university (30%). Panel (b) reports what mentors perceived as the importance of different factors to their mentee’s university application decisions.⁴⁰ The factor with the highest average importance reported was whether students would fit in and enjoy their time at university, with a mean importance of 3.8, and in a majority of conversations (56%) mentors reported a weakly higher importance score for this factor than any other factor. Prospects of academic success on the course, careers, prospects of getting an offer and course content were all reported as less important.

This is consistent with the results from Section B illustrating that video exposure affects social beliefs more than offer or graduation beliefs; both sets of results indicate that the social environment at university is the primary topic that students seek to learn about from interactions with past students.

VII. Conclusion

Socioeconomically disadvantaged students enroll at elite universities at lower rates compared to high-SES students with similar academic backgrounds. Widespread efforts to address these disparities typically involve increasing transparency on admission requirements or directly lowering admission thresholds for low-SES students. But much of what students care about when making application choices, such as whether *someone like them would fit in*, can be learned only through direct contact with past attendees. Students at socioeconomically disadvantaged schools are less likely to interact with schoolmates who go on to attend prestigious universities. Our paper illustrates how this lack of exposure can deter subsequent cohorts of students from applying to elite universities, dynamically reinforcing this gap. Our back-of-envelope calculation attributes 15–30% of the socioeconomic gap in elite university applications to differences in exposure.

students need to actually ask mentors about a topic to learn relevant information. This avoids potential demand effects that would arise if we directly ask students what they are considering in their university applications, as students may think they are ‘supposed’ to decide based on factors like course content and teaching quality that are usually reported in university guides, and respond with these answers.

39. We ask mentors to select the topics from a multiple-choice list that they discussed with each of their mentees; the topics selected are reported in panel (a).

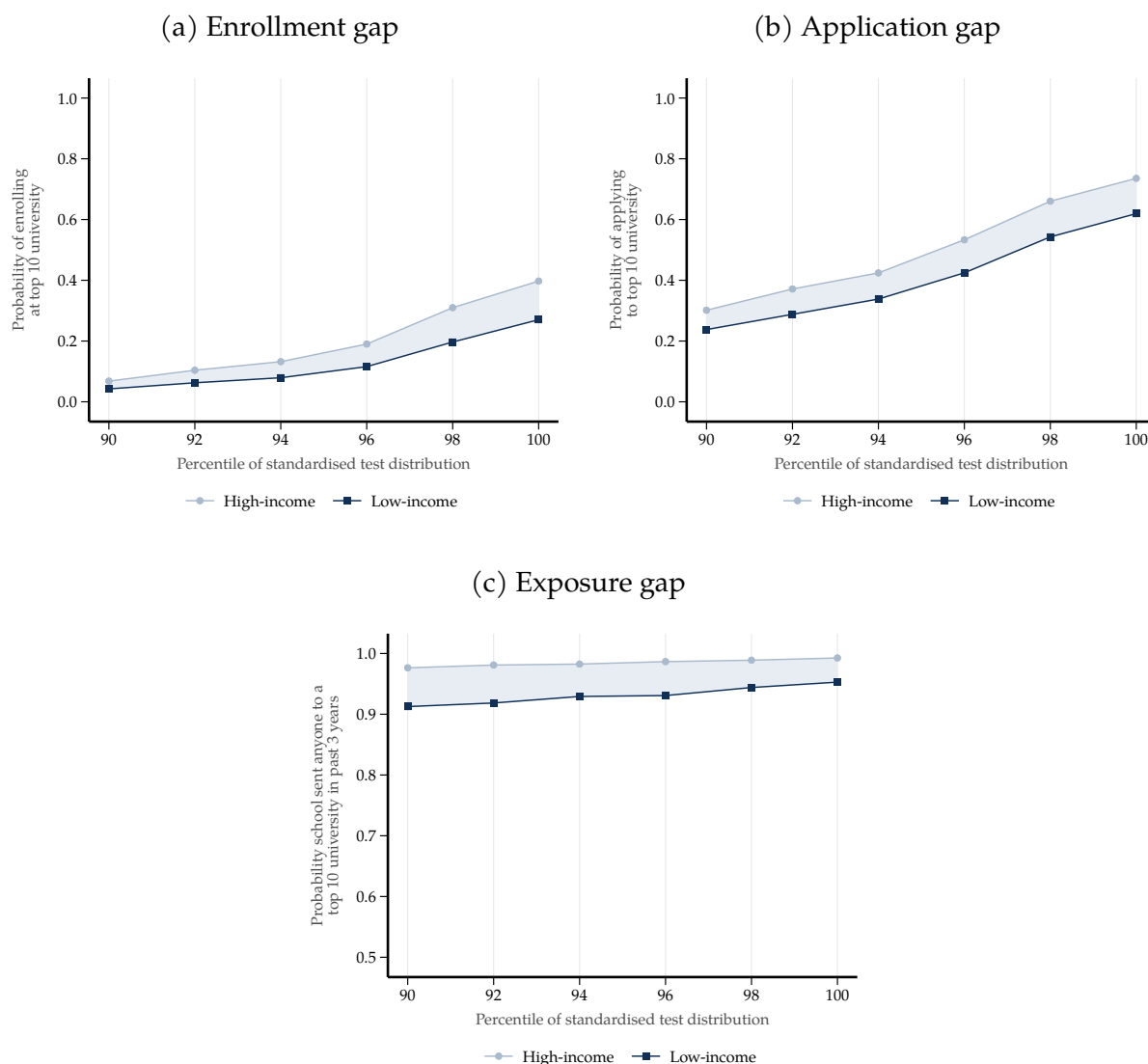
40. For each of the factors listed in panel (b), mentors reported importance on a 5-point scale from ‘Not at all important’ to ‘extremely important’.

A student who *breaks through* to an elite university, however, can break this cycle by conveying tacit information that is both demographic group-specific and university-specific. Our surveys with prospective university applicants indicate that low-SES students are pessimistic about their chances of fitting in at elite universities, but are neither miscalibrated on the grades required for admission nor pessimistic about their chances of admission or graduation. Our experiment, which randomizes exposure to elite universities outside of participants' existing social networks, shifts beliefs about social fit, but has no effect on admissions or graduation beliefs. Consistent with our findings in the administrative data, experimentally induced exposure increased applications to treatment universities with particularly strong effects when participants and mentors were demographically similar.

Low-SES students in our experiment are *more responsive* to elite exposure than high-SES students. Yet the presence of more high-SES classmates can generate negative relative-rank effects that offset increases in university aspirations among low-SES students (Cattan, Salvanes, and Tominey 2025). By offering social connectedness without class-rank effects, our experiment highlights external mentorship and social exposure as a promising avenue for addressing inequality of opportunity across socioeconomically stratified school networks. From a policy perspective, our results also suggest that lowering admissions requirements for disadvantaged students (e.g., affirmative action in the US and contextual offers in the UK) may be insufficient to close socioeconomic gaps at elite universities. At the same time, policies that require universities to admit high-achieving students from a wider range of schools, such as the Top 10 Percent Rule in Texas, may have positive spillover effects on the applications from future cohorts. Quantifying the implications of these dynamic spillovers for optimal university admissions policy is an important avenue for future research.

VIII. Exhibits

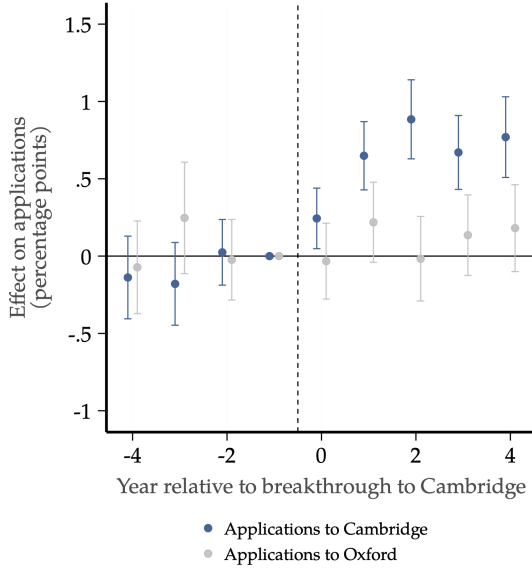
Figure 1: Socioeconomic gaps in enrollment, applications, and exposure at elite universities



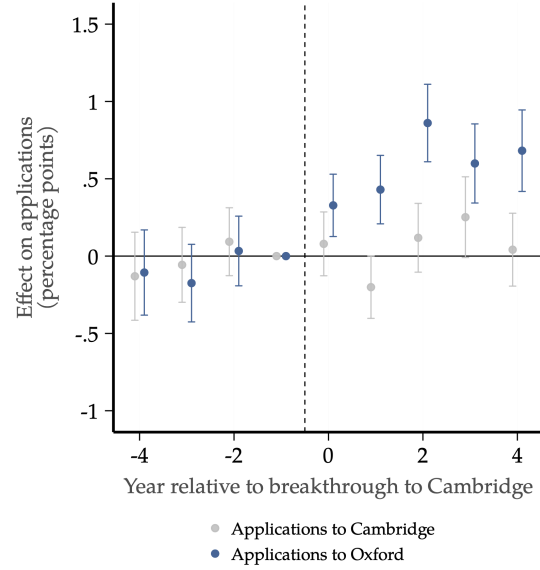
Notes: Panels (a) and (b) plot the probability of enrolling at (panel (a)) and applying to (panel (b)) a top 10 university, separately by parental income, across the distribution of standardized test scores (GCSE percentile). Low-income students are students in the bottom two quintiles of those whose home postcode is in the bottom 40% of neighbourhoods as ranked by the Index of Multiple Deprivation (IMD); high-income students are students in the top two quintiles of IMD. Panel (c) plots the corresponding gap in exposure to top 10 universities, defined the probability of attending a school where at least one student attending a top 10 university in the preceding three years.

Figure 2: Effects of exposure to breakthroughs on university outcomes

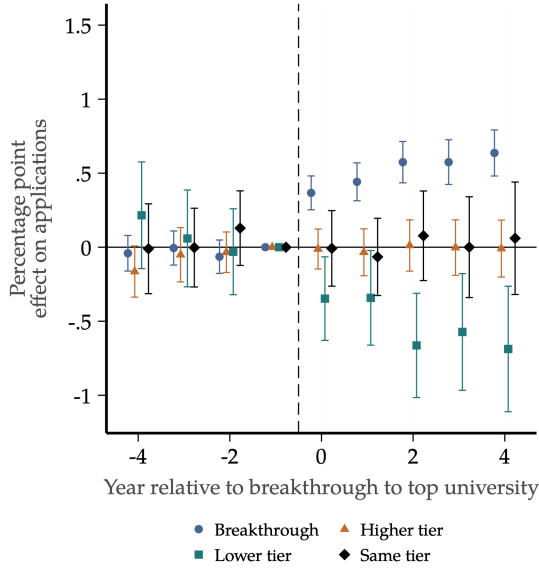
(a) Applications, breakthrough to Cambridge



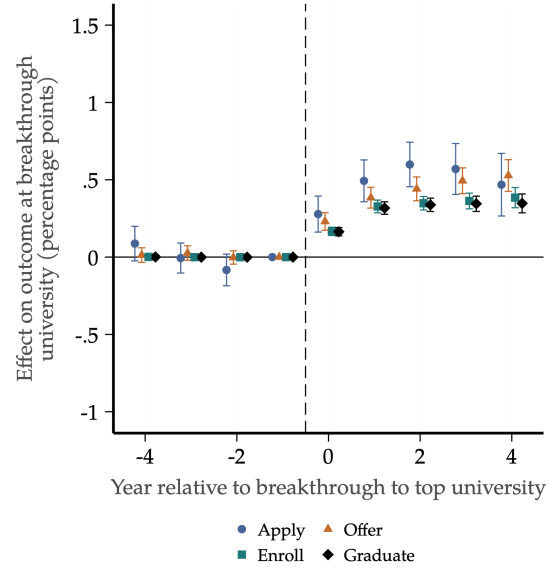
(b) Applications, breakthrough to Oxford



(c) Application portfolio outcomes



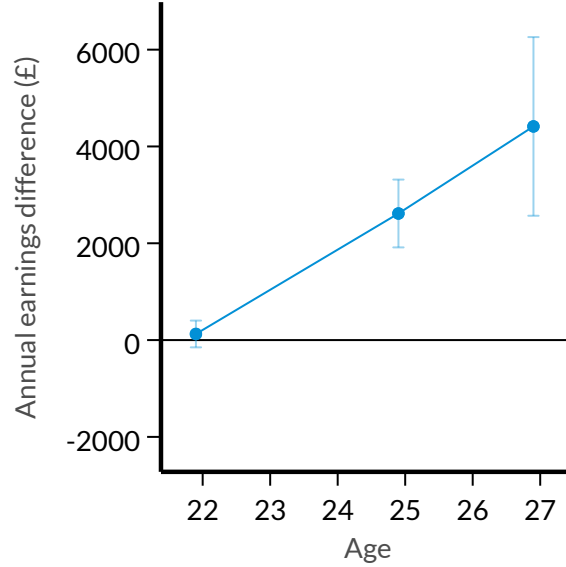
(d) Offer, enrollment and graduation outcomes



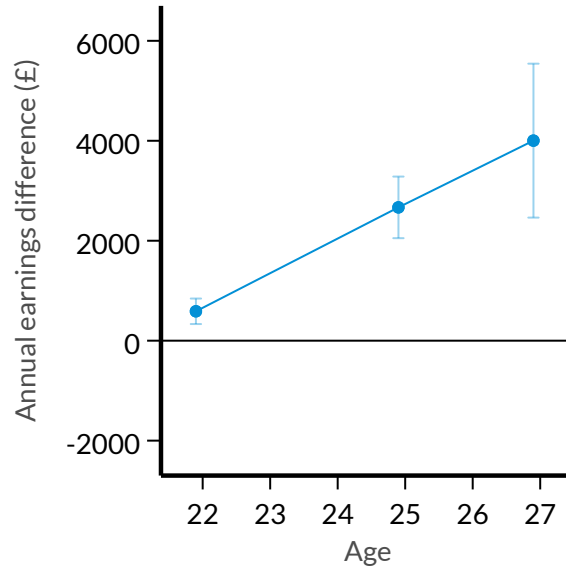
Notes: Panels (a) and (b) estimate equation 5 for breakthroughs to Cambridge and to Oxford respectively, where the outcome is applications to that university; these regressions include school and year fixed effects. Panels (c) and (d) estimate equation 6, pooling across top 10 universities (top 5 in c), and include school-university and year-university fixed effects. Panel (c) reports four mutually exclusive application portfolio outcomes: the breakthrough university; a university in the same tier; a university in a higher tier; and a university in a lower tier. Panel (d) reports indicators for applying to, receiving an offer from, enrolling at, and graduating from the breakthrough university; cohorts after 2016 are excluded from panel (d) as graduation is not observed for these cohorts. Universities are ranked by the mean A-level grades of their enrolled students. We estimate breakthrough effects using the Sun and Abraham estimator and multiply coefficients by 100 so they can be interpreted in percentage point terms. 95% confidence intervals reported based on standard errors clustered at the school level.

Figure 3: Labor market outcomes for students induced into elite universities

(a) Earnings effects relative to matched control schools

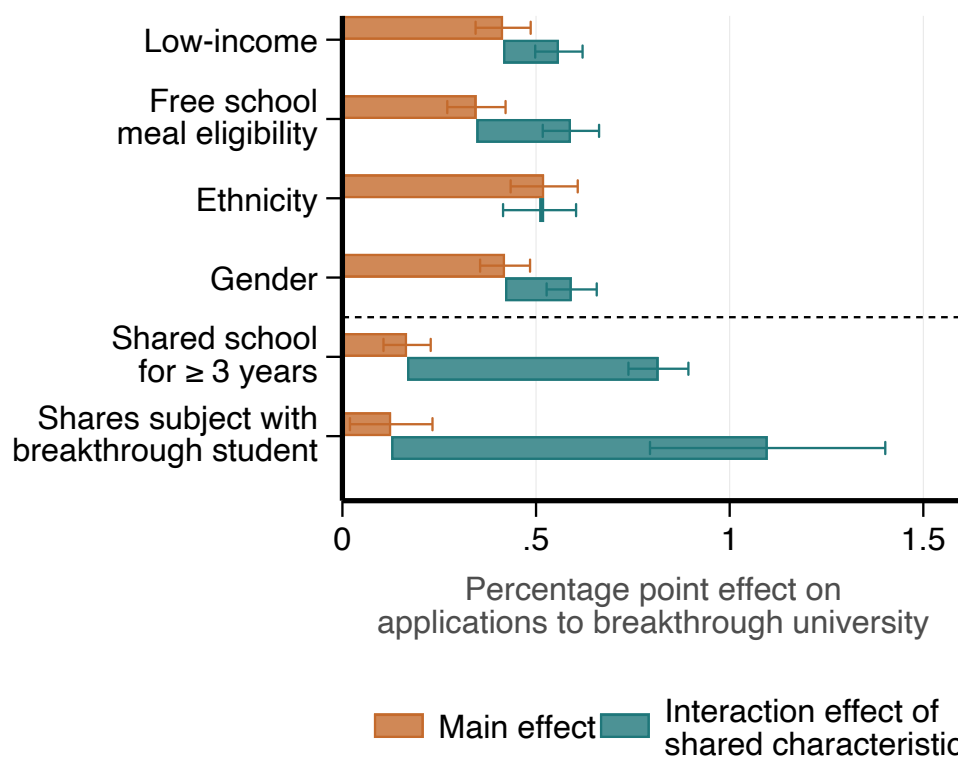


(b) Earnings effects relative to matched control cohorts



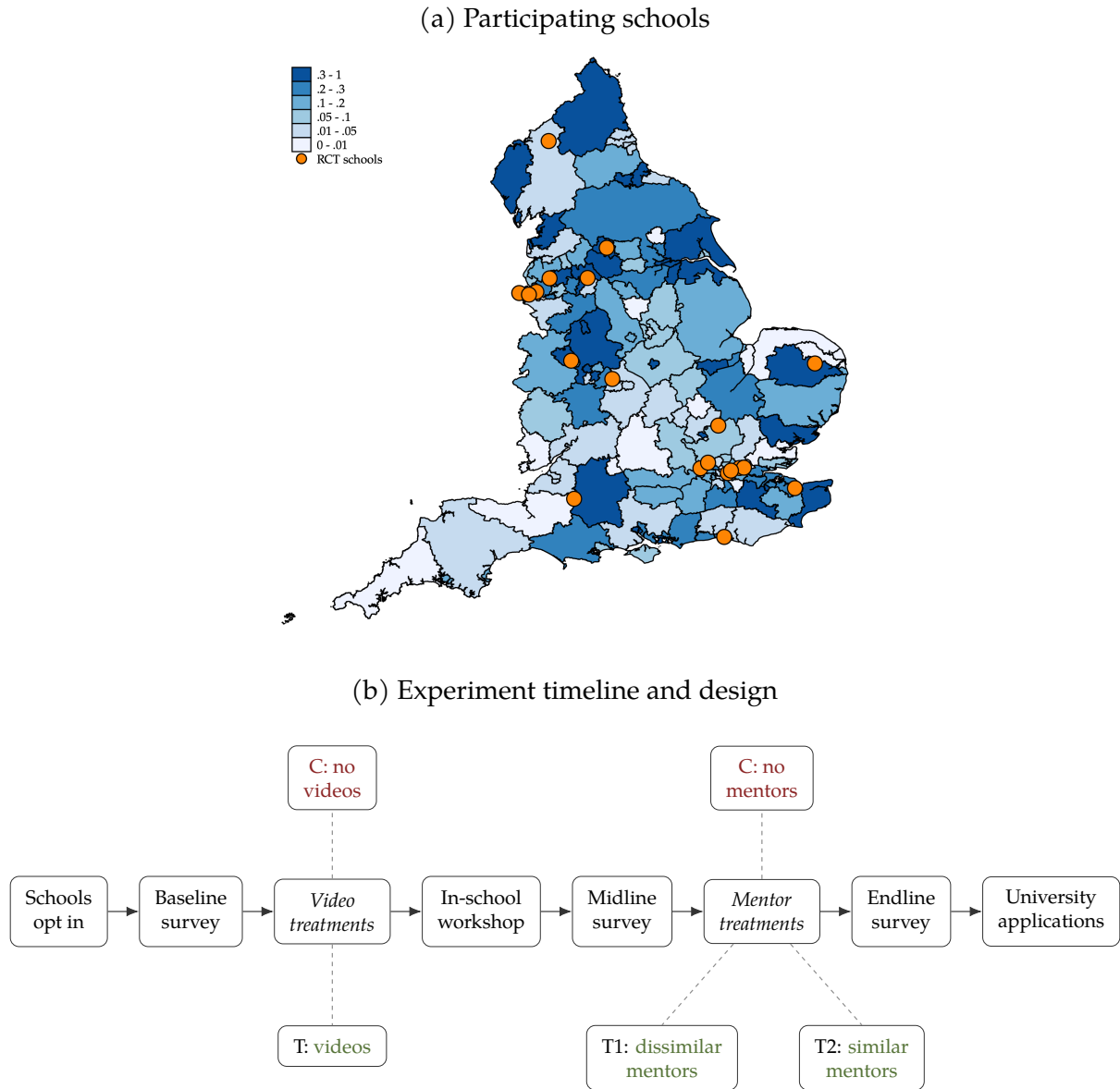
Notes: 95% confidence intervals throughout, based on standard errors clustered at the school level. Panels A and B report regressions of earnings at the specified age on an indicator for applying to the breakthrough university following a breakthrough, controlling for matched pair effects. Matched control students are drawn from a control set consisting of students applying for the same major in the same year at control high schools in panel (a), and of students at the same school prior to treatment in panel (b). Within the control set for each student, we also exactly match on the student's ventile in the sample GCSE grade distribution and an indicator for whether the student is taking 3 or more A-levels, and then select one nearest neighbour by Mahalanobis distance, matching on gender, neighbourhood income decile, and GCSE grades.

Figure 4: Heterogeneity in the effect of breakthroughs on applications, by similarity between the student and the breakthrough student



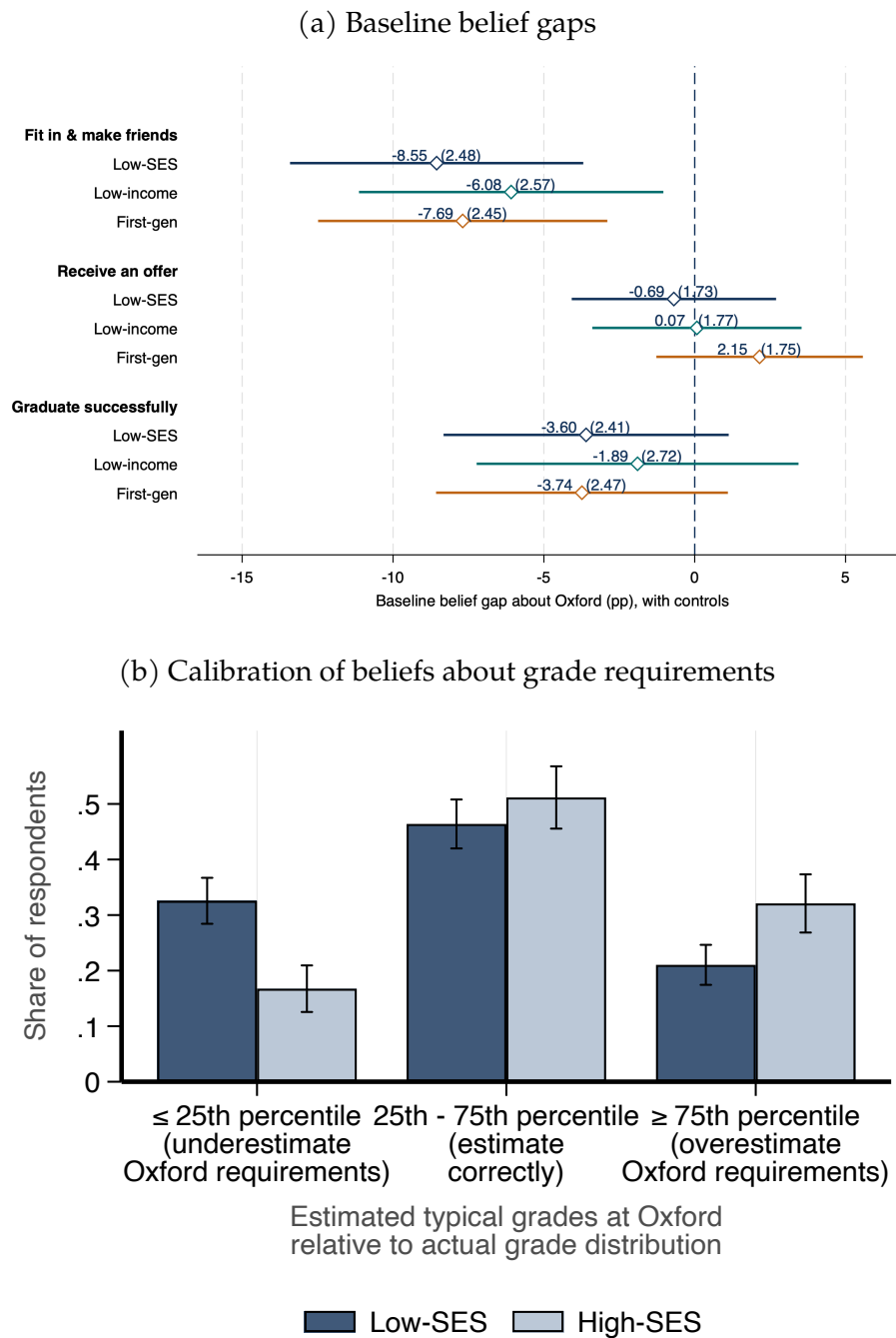
Notes: Coefficients from difference-in-difference regressions estimating the effect of breakthroughs on applications to the breakthrough university, pooled across breakthroughs to top 10 universities (with the exception of the shares subject regression, for which we pool across the top 7). For each characteristic listed on the vertical axis, we interact the Treated $\times$ Post indicator with an indicator for whether the student shares that characteristic with the breakthrough student at their school. The blue bars report the main Treated $\times$ Post effect for students who do not share the characteristic, and the red bars the additional effect for students who do. 'Shared school for ≥ 3 years' denotes students who attended the same age-16 school as the breakthrough student (by construction, as students are compared to a breakthrough student from the school they attended at age 18, these students share a school for at least 3 years). Coefficients are multiplied by 100 so they can be interpreted in percentage point terms. Regressions include school-by-breakthrough-university and year-by-breakthrough-university fixed effects. 95% confidence intervals reported based on standard errors clustered at the school level.

Figure 5: Overview of Experimental Design and Sample



Notes: Panel (a) maps the locations of schools participating in our RCT within England; each orange dot represents a participating school. The underlying map plots the share of students in each ITL 3 region of England who attended high schools that sent no one to a top 10 university in the preceding three years, as of 2018. One participating school in Wales is omitted from the map. Panel (b) outlines the stages of the RCT and the treatment arms, as implemented in the Spring 2025 wave.

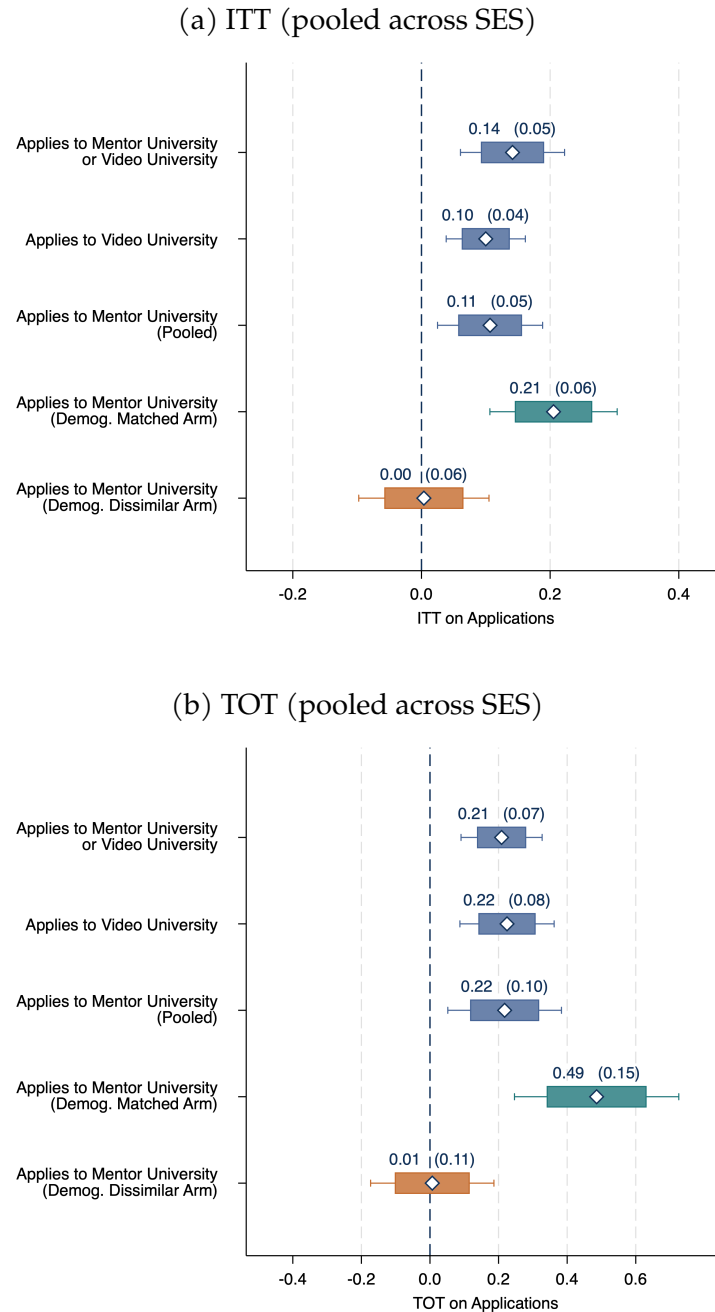
Figure 6: Baseline beliefs about elite universities



Notes: Data: RCT. Sample restricted to students who completed both baseline and midline surveys. Beliefs are subjective probabilities, between 0 and 100, of 'receiving an offer from', 'fitting in and making friends at', or 'graduating successfully from' Oxford University. Low-income students are those whose home postcode is in the bottom 40% of neighbourhood deprivation. Low-SES students are students who are *either* low-income or whose parents did not attend university.

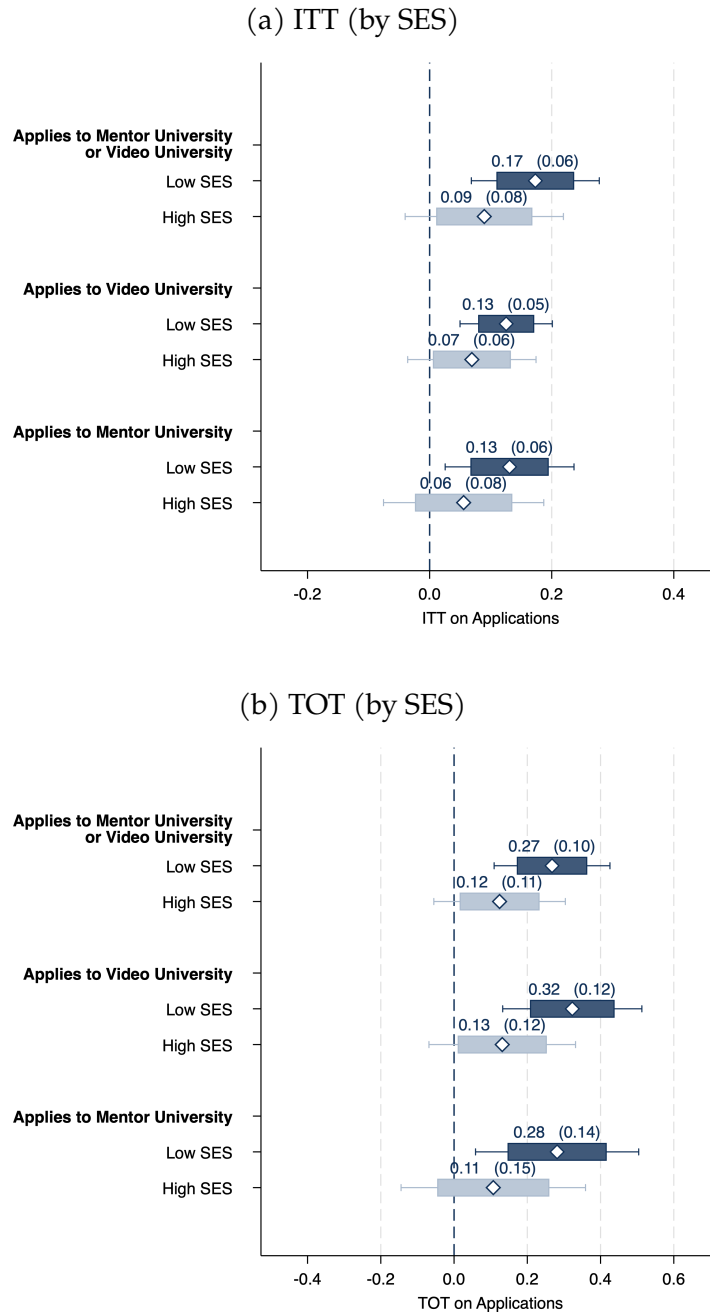
Panel (a): coefficients from regressions of each belief on the specified student characteristic and a control for the student's predicted A-level grades, reported as the average percentage point difference in probability relative to other students; 90% confidence intervals. Panel (b): the share of respondents whose estimate of typical grades at Oxford falls below the 25th percentile, between the 25th and 75th percentile, or above the 75th percentile of the actual grade distribution of enrolled students, separately for low- and high-income students; 90% confidence intervals.

Figure 7: Treatment effects on university applications



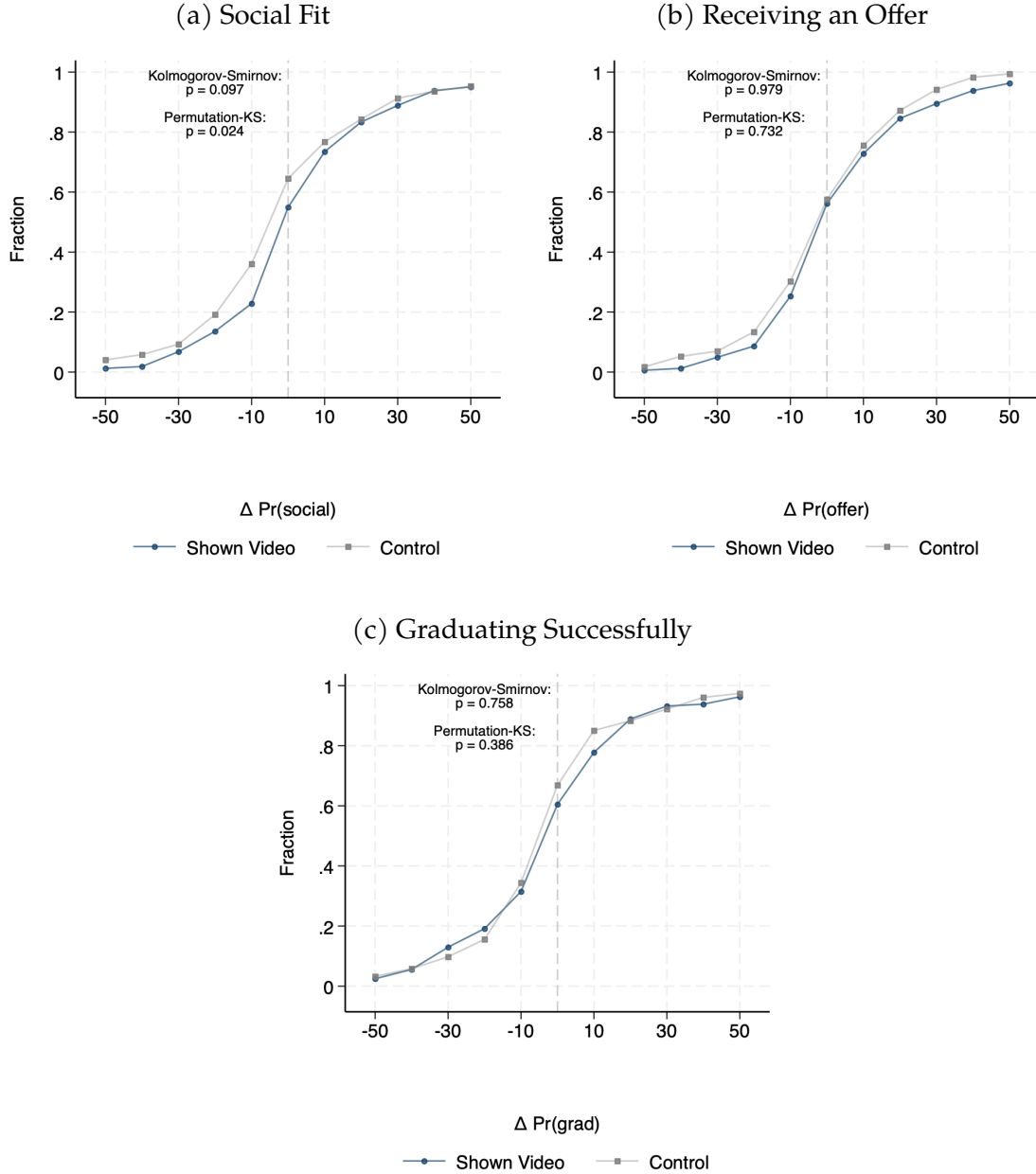
Notes: Treatment effects of videos and mentors on applying to the universities discussed. ‘Mentor university’ is defined as one of the 1–2 universities that a student’s assigned mentors attended, and ‘video university’ is defined as one of the 2 universities that the speakers in a student’s assigned videos attended. The outcome in each case is a binary indicator for applying to at least one of the specified universities. We regress the specified application outcome on either the pooled treatment arm assignment T (in the first three rows of each panel) or the demographic match arms T_1 and T_2 (the demog. dissimilar and demog. matched arms respectively), in each case relative to the control group C . In panel (a), we report intent-to-treat estimates, and in panel (b) we instrument for takeup with the treatment arm dummies and identify the treatment-on-treated effects. The sample is restricted to students who responded ‘Maybe’, ‘Probably’, or ‘Definitely’ when asked whether they would be interested in a mentor.

Figure 8: Treatment effects on university applications, by SES



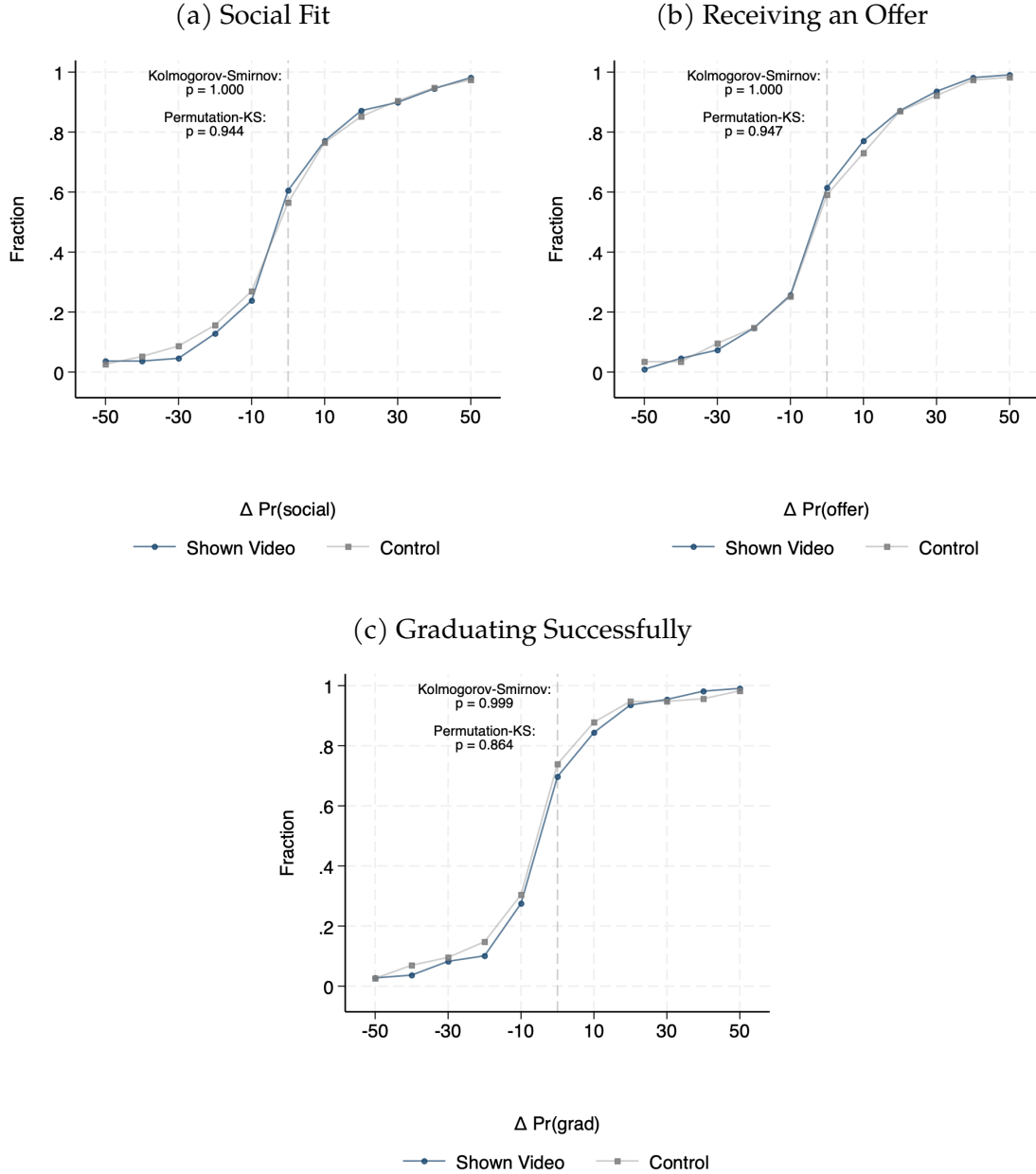
Notes: Treatment effects of videos and mentors on applying to the universities discussed. ‘Mentor university’ is defined as one of the 1–2 universities that a student’s assigned mentors attended, and ‘video university’ is defined as one of the 2 universities that the speakers in a student’s assigned videos attended. The outcome in each case is a binary indicator for applying to at least one of the specified universities. Low-SES students are those who either have a home in the bottom 40% of neighborhood deprivation or who have no parents who have attended university; all others are high-SES. We regress the specified application outcome on either the pooled treatment arm assignment T , with the sample restricted to the specified SES group. In panel (a), we report intent-to-treat estimates, and in panel (b) we instrument for takeup with the treatment arms and identify the treatment-on-treated effects. The sample is restricted to students who responded ‘Maybe’, ‘Probably’, or ‘Definitely’ when asked whether they would be interested in a mentor.

Figure 9: Treatment effects on beliefs about video university, low-SES



Notes: Sample restricted to students who completed both baseline and midline surveys. Each panel plots the cumulative distribution of the percentage point change between the baseline and midline survey in the reported probability of the specified outcome at the student's assigned video university, separately for students who were and were not assigned to the video treatment arms. We pool across arms T1 and T2. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of the two distributions. Elicited beliefs were restricted to be multiples of 10, so distributions are discrete.

Figure 10: Treatment effects on beliefs about video university, high-SES



Notes: Sample restricted to students who completed both baseline and midline surveys. Each panel plots the cumulative distribution of the percentage point change between the baseline and midline survey in the reported probability of the specified outcome at the student's assigned video university, separately for students who were and were not assigned to the video treatment arms. We pool across arms T1 and T2. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of the two distributions. Elicited beliefs were restricted to be multiples of 10, so distributions are discrete.

Table 1: Summary statistics

				Experimental sample	
	(1)	(2)	(3)	(4)	(5)
	Age-18 school leavers	University applicants	Event study sample	Full sample	Endline observed
<i>Panel A. Demographic characteristics</i>					
Female	51.2	56.0	57.3	42.2	43.8
Low-income	37.2	28.2	26.3	35.7	37.3
Free school meal eligible	21.6	12.5	11.6	—	—
Parents attended university	—	56.0 [†]	57.7 [†]	54.5	53.5
White	80.0	80.6	81.1	56.7	53.7
Black	4.6	4.8	4.5	6.9	7.6
Asian	9.2	10.2	10.2	24.8	28.3
Other	6.3	4.4	4.2	11.6	10.4
<i>Panel B. School region</i>					
Northern England	28.2	28.7	28.9	42.8	37.6
The Midlands	19.6	19.3	19.2	3.0	1.5
Southern England	38.0	36.9	36.8	33.1	36.2
London	14.1	15.1	15.1	20.8	24.5
<i>Panel C. Academic characteristics</i>					
Taking ≥ 3 A-levels	35.2	66.6	68.9	85.0	85.6
Achieved tariff points (med.)	104	112	112	—	—
Predicted tariff points	—	—	—	126.4	125.1
N	7,164,386	2,920,445	2,290,950	640	473

Notes: All entries are percentages unless otherwise noted. The event study sample is students at schools that either experience a breakthrough or are a control for one. The RCT sample is students completing both the baseline and midline survey. Low-income means a home postcode in the bottom 40% of the Index of Multiple Deprivation. A-level tariff points convert letter grades to a 0–56 scale, summed over a student’s top 3 grades (maximum 168). Column (5) reports the control group mean and column (6) the coefficient on an indicator for assignment to a mentor treatment arm, both in percentage points, with heteroskedasticity-robust standard errors in parentheses.

[†] Parental university attendance in LEO is observed only for students who themselves attend university.

Table 2: Mentor conversation topic reports

(a) Topics Discussed in Mentor-Participant Conversations

	Conversations where topic was discussed (%)
Student Life at Mentor's University	70
Life at University and Fitting In	68
Mentor's Course (university-major)	66
Application Advice (e.g., Personal Statement)	64
Choice of Course (university-major)	53
Post-degree Career	36
Academic Success at University	30
<i>N</i>	103

(b) Mentor-reported importance of choice factors for students

	Importance (1–5 scale)		Z-score	Most important (%) [†]
	Mean	SE		
Whether they would fit in and enjoy their time	3.80	0.10	0.18	56.7
Whether they would succeed academically	3.65	0.11	0.03	35.1
Jobs that the course could help them to get	3.59	0.11	-0.03	38.1
Whether they could get an offer from the course	3.58	0.11	-0.04	43.3
Course content and teaching quality	3.46	0.10	-0.16	29.9
<i>N</i>	103	103	103	103

Notes: Descriptive statistics on topics discussed in calls and emails with mentors, based on the post-mentorship survey. In panel (a), mentors were asked a multiple choice question to select all topics they had discussed with each of their mentees, from the topics listed in the table; we tabulate the share of conversations for which the specified topic was selected. In panel (b), mentors were asked how important each of the listed factors was to the student they met with as the student decided which course to apply to, on a qualitative scale from “Not at all important” to “Extremely important”. We code these responses onto a 1–5 scale (where 5 is “extremely important”) and report the mean response, standard error, Z-score (based on the pooled mean and standard deviation of importance scores across factors), and the share of students for which each factor was (possibly jointly) most important according to their mentor.

[†] Ties were allowed when ranking importance, so shares do not sum to 1.

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